$$= 1 - \frac{1}{\left(r_p^{\frac{1}{\gamma}}\right)^{\gamma-1}} = 1 - \frac{1}{r_p^{\left(\frac{\gamma-1}{\gamma}\right)}} \qquad (2.81)$$

From Eq.2.81, it is seen that the efficiency of the Brayton cycle depends only on the pressure ratio and the ratio of specific heat, γ.

$$\begin{aligned}
\text{Network output} &= \text{Expansion work} - \text{Compression work} \\
&= C_p(T_3 - T_4) - C_p(T_2 - T_1) \\
&= C_pT_1\left(\frac{T_3}{T_1} - \frac{T_4}{T_1} - \frac{T_2}{T_1} + 1\right) \\
\frac{W}{C_pT_1} &= \frac{T_3}{T_1} - \frac{T_4}{T_3}\frac{T_3}{T_1} - \frac{T_2}{T_1} + 1 \qquad (2.82)
\end{aligned}$$

It can be easily seen from the Eq.2.82 (work output) that, the work output of the cycle depends on initial temperature, T_1, the ratio of the maximum to minimum temperature, $\frac{T_3}{T_1}$, pressure ratio, r_p and γ which are used in the calculation of $\frac{T_2}{T_1}$. Therefore, for the same pressure ratio and initial conditions work output depends on the maximum temperature of the cycle.

Worked out Examples

OTTO CYCLE

2.1 An engine working on Otto cycle has the following conditions : Pressure at the beginning of compression is 1 bar and pressure at the end of compression is 11 bar. Calculate the compression ratio and air-standard efficiency of the engine. Assume $\gamma = 1.4$.

Solution

$$r = \frac{V_1}{V_2} = \left(\frac{p_2}{p_1}\right)^{\left(\frac{1}{\gamma}\right)} = 11^{\frac{1}{1.4}} = 5.54 \quad \underleftarrow{\text{Ans}}$$

$$\eta_{\text{air-std}} = 1 - \frac{1}{r^{\gamma-1}} = 1 - \left(\frac{1}{5.54}\right)^{0.4}$$

$$= 0.496 = 49.6\% \quad \underleftarrow{\text{Ans}}$$

2.2 In an engine working on ideal Otto cycle the temperatures at the beginning and end of compression are 50 °C and 373 °C. Find the compression ratio and the air-standard efficiency of the engine.

Solution

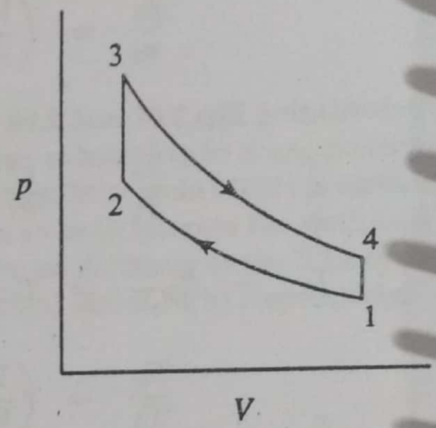
$$r = \frac{V_1}{V_2} = \left(\frac{T_1}{T_2}\right)^{\frac{1}{\gamma-1}} = \left(\frac{646}{323}\right)^{\frac{1}{0.4}} = 5.66 \quad \Leftarrow \text{Ans}$$

$$\eta_{Otto} = 1 - \frac{1}{r^{\gamma-1}} = 1 - \frac{T_1}{T_2}$$

$$= 1 - \frac{323}{646} = 0.5 = 50\% \quad \Leftarrow \text{Ans}$$

2.3 In an Otto cycle air at 17 °C and 1 bar is compressed adiabatically until the pressure is 15 bar. Heat is added at constant volume until the pressure rises to 40 bar. Calculate the air-standard efficiency, the compression ratio and the mean effective pressure for the cycle. Assume $C_v = 0.717$ kJ/kg K and $R = 8.314$ kJ/kmol K.

Solution



Consider the process 1 – 2

$$p_1 V_1^{\gamma} = p_2 V_2^{\gamma}$$

$$\frac{V_1}{V_2} = r = \left(\frac{p_2}{p_1}\right)^{\frac{1}{\gamma}}$$

$$= \left(\frac{15}{1}\right)^{\frac{1}{1.4}} = 6.91$$

$$\eta = 1 - \left(\frac{1}{r}\right)^{\gamma-1}$$

$$= 1 - \left(\frac{1}{6.91}\right)^{0.4} = 0.539 = 53.9\%$$

$$T_2 = \frac{p_2 V_2}{p_1 V_1} T_1 = \frac{15}{1} \times \frac{1}{6.91} \times 290 = 629.5 \text{ K}$$

Consider the process 2 – 3

$$T_3 = \frac{p_3 T_2}{p_2} = \frac{40}{15} \times 629.5 = 1678.7$$

$$\textit{Heat supplied} = C_v(T_3 - T_2)$$

$$= 0.717 \times (1678.7 - 629.5) = 752.3 \text{ kJ/kg}$$

$$\textit{Work done} = \eta \times q_s = 0.539 \times 752.3 = 405.5 \text{ kJ/kg}$$

$$p_m = \frac{\text{Work done}}{\text{Swept volume}}$$

$$v_1 = \frac{V_1}{m} = M\frac{RT_1}{p_1} = \frac{8314 \times 290}{29 \times 1 \times 10^5} = 0.8314 \text{ m}^3/\text{kg}$$

$$v_1 - v_2 = \frac{5.91}{6.91} \times 0.8314 = 0.711 \text{ m}^3/\text{kg}$$

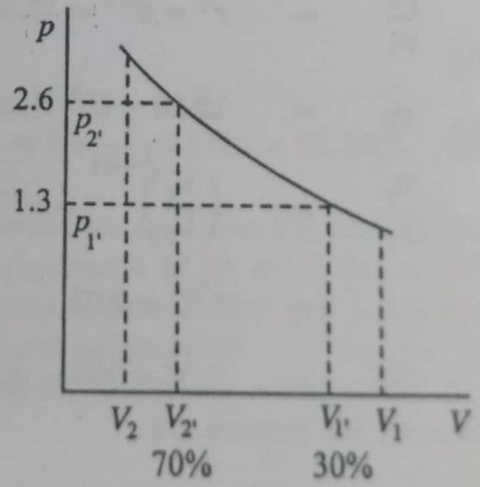
$$p_m = \frac{405.5}{0.711} \times 10^3 = 5.70 \times 10^5 \text{ N/m}^2$$

$$= 5.70 \text{ bar}$$

Ans

2.4 Fuel supplied to an SI engine has a calorific value 42000 kJ/kg. The pressure in the cylinder at 30% and 70% of the compression stroke are 1.3 bar and 2.6 bar respectively. Assuming that the compression follows the law $pV^{1.3}$ = constant. Find the compression ratio. If the relative efficiency of the engine compared with the air-standard efficiency is 50%. Calculate the fuel consumption in kg/kW h.

Solution



$$V_2 = 1$$

$$V_{1'} = 1 + 0.7(r - 1)$$

$$= 0.7r + 0.3$$

$$V_{2'} = 1 + 0.3(r - 1)$$

$$= 0.3r + 0.7$$

$$\frac{V_{1'}}{V_{2'}} = \left(\frac{p_2}{p_1}\right)^{\frac{1}{n}}$$

$$= \left(\frac{2.6}{1.3}\right)^{\frac{1}{1.3}} = 1.7$$

$$\frac{0.7r + 0.3}{0.3r + 0.7} = 1.7$$

$$r = 4.68$$

Ans

$$\eta_{air-std} = 1 - \frac{1}{r^{\gamma - 1}}$$

$$= 1 - \frac{1}{4.68^{0.4}} = 0.46 = 46\%$$

$$\text{Relative efficiency} = \frac{\text{Indicated thermal efficiency}}{\text{Air-standard efficiency}}$$

$$\eta_{ith} = 0.5 \times 0.46 = 0.23$$

$$\eta_{th} = \frac{ip}{CV \times \dot{m}}$$

where $\dot{m}$ is in kg/s

$$\frac{\dot{m}}{ip} = \frac{1}{42000 \times 0.23}$$

$$= 1.035 \times 10^{-4} \text{ kg/kW s}$$

$$= 1.035 \times 10^{-4} \times 3600 \text{ kg/kW h}$$

$$isfc = 0.373 \text{ kg/kW h}$$ Ans

2.5 A gas engine working on the Otto cycle has a cylinder of diameter 200 mm and stroke 250 mm. The clearance volume is 1570 cc. Find the air-standard efficiency. Assume $C_p = 1.004$ kJ/kg K and $C_v = 0.717$ kJ/kg K for air.

Solution

$$V_s = \frac{\pi}{4} d^2 L = \frac{\pi}{4} \times 20^2 \times 25 = 7853.98 \text{ cc}$$

$$r = 1 + \frac{V_s}{V_c} = 1 + \frac{7853.98}{1570} = 6.00$$

$$\gamma = \frac{C_p}{C_v} = \frac{1.004}{0.717} = 1.4$$

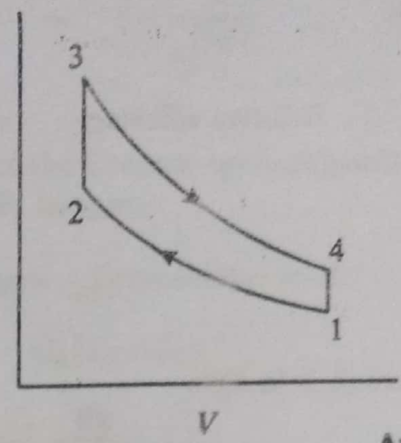
$$\eta_{\text{air-std}} = 1 - \frac{1}{r^{\gamma-1}} = 1 - \frac{1}{6^{(0.4)}} \times 100 = 51.2\%$$ Ans

2.6 In a S.I. engine working on the ideal Otto cycle, the compression ratio is 5.5. The pressure and temperature at the beginning of compression are 1 bar and 27 °C respectively. The peak pressure is 30 bar. Determine the pressure and temperatures at the salient points, the air-standard efficiency and the mean effective pressure. Assume ratio of specific heats to be 1.4 for air.

Solution

Since $V_2 = V_3 = V_c$

$V_1 = rV_2 = rV_c,$



Consider the process 1 – 2,

$$\frac{p_2}{p_1} = r^{\gamma} = 5.5^{1.4} = 10.88$$

$$p_2 = 10.88 \times 1 \times 10^5$$

$$= 10.88 \times 10^5\ \text{N/m}^2$$

$$\frac{T_2}{T_1} = r^{\gamma-1} = 5.5^{0.4} = 1.978$$

$$T_2 = 1.978 \times 300 = 593.4\ \text{K} = 320.4°\ \text{C}$$ Ans

Consider the process 2 – 3,

$$p_3 = 30 \times 10^5\ \text{N/m}^2$$

$$\frac{T_3}{T_2} = \frac{p_3}{p_2} = \frac{30}{10.88} = 2.757$$

$$T_3 = 2.757 \times 593.4 = 1636\ \text{K}$$

$$= 1363°\ \text{C}$$ Ans

Consider the process 3 – 4,

$$\frac{p_3}{p_4} = \left(\frac{V_4}{V_3}\right)^{\gamma} = \left(\frac{V_1}{V_2}\right)^{\gamma}$$

$$= r^{\gamma} = 5.5^{1.4} = 10.88$$

$$p_4 = \frac{p_3}{10.88} = 2.76 \times 10^5\ \text{N/m}^2$$ Ans

$$\frac{T_3}{T_4} = r^{\gamma-1} = 5.5^{0.4} = 1.978$$

$$T_4 = \frac{T_3}{1.978} = \frac{1636}{1.978} = 827.1\ \text{K} = 554.1°\ \text{C}$$ Ans

$$\eta_{Otto} = 1 - \frac{1}{r^{(\gamma-1)}} = 1 - \frac{1}{5.5^{0.4}} = 0.4943 = 49.43\%$$ Ans

$$p_m = \frac{\text{Indicated work/cycle}}{V_s} = \frac{\text{Area of p-V diagram 1234}}{V_s}$$

$$\text{Area 1234} = \text{Area under 3-4} - \text{Area under 2-1}$$

$$= \frac{p_3V_3 - p_4V_4}{\gamma - 1} - \frac{p_2V_2 - p_1V_1}{\gamma - 1}$$

$$= \quad 23.6 \times 10^5 \times V_c = p_m \times V_s$$

$$p_m = \frac{23.6 \times 10^5 \times V_c}{V_s} = \frac{23.6 \times 10^5 \times V_c}{4.5 \times V_c}$$

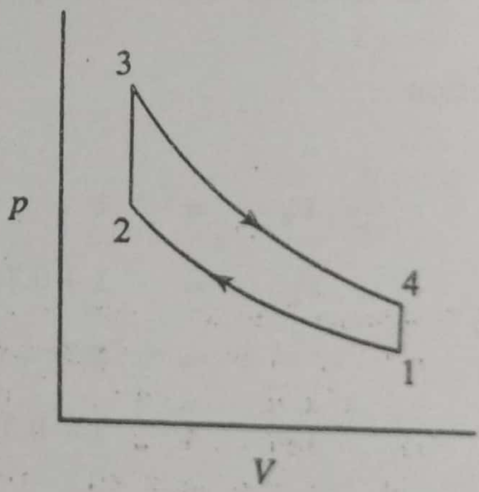
$$= \quad 5.24 \times 10^5 \text{ N/m}^2 = \mathbf{5.24 \text{ bar}}$$

Ans $\Leftarrow$

2.7 A gas engine operating on the ideal Otto cycle has a compression ratio of 6:1. The pressure and temperature at the commencement of compression are 1 bar and 27 °C. Heat added during the constant volume combustion process is 1170 kJ/kg. Determine the peak pressure and temperature, work output per kg of air and air-standard efficiency. Assume C_v = 0.717 kJ/kg K and γ = 1.4 for air.

Solution

Consider the process 1-2



$$\frac{p_2}{p_1} = r^\gamma = 6^{1.4} = 12.28$$

$$p_2 = 12.28 \times 10^5 \text{ N/m}^2$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = r^{\gamma-1}$$

$$= 6^{0.4} = 2.05$$

$$T_2 = 2.05 \times 300 = 615K = 342^\circ \text{ C}$$

Consider the process 2-3

For unit mass flow

$$q_s = q_{2-3} = C_v(T_3 - T_2) = 1170 \text{ kJ/kg}$$

$$T_3 - T_2 = \frac{1170}{0.717} = 1631.8$$

$$T_3 = 1631.8 + 615 = 2246.8 \text{ K}$$

$$= \mathbf{1973.8^\circ \text{ C}}$$

Ans $\Leftarrow$

$$\frac{p_3}{p_2} = \frac{T_3}{T_2} = \frac{2246.8}{615} = 3.65$$

$$\text{Peak pressure}, p_3 = 3.65 \times 12.28 \times 10^5$$

$$= 44.82 \times 10^5 \text{ N/m}^2 = \mathbf{44.82\ bar} \quad \underleftarrow{\text{Ans}}$$

$$\text{Work output} = \text{Area of } p\text{-}V \text{ diagram}$$

$$= \text{Area under } (3 - 4) - \text{Area under } (2 - 1)$$

$$= \frac{p_3V_3 - p_4V_4}{\gamma - 1} - \frac{p_2V_2 - p_1V_1}{\gamma - 1}$$

$$= \frac{mR}{\gamma - 1}[(T_3 - T_4) - (T_2 - T_1)]$$

$$R = C_p - C_v = 1.004 - 0.717$$

$$= 0.287 \text{ kJ/kg K}$$

$$\frac{T_3}{T_4} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} = r^{(\gamma-1)} = 6^{0.4} = 2.048$$

$$T_4 = \frac{T_3}{2.048} = \frac{2246.8}{2.048} = 1097.1 \text{ K}$$

$$\text{Work output/kg} = \frac{0.287}{0.4} \times [(2246.8 - 1097.1) - (615 - 300)]$$

$$= \mathbf{598.9\ kJ} \quad \underleftarrow{\text{Ans}}$$

$$\eta_{Otto} = 1 - \frac{1}{r^{(\gamma-1)}} = 1 - \frac{1}{6^{0.4}} 0.5116 = \mathbf{51.16\%} \quad \underleftarrow{\text{Ans}}$$

2.8 A spark-ignition engine working on ideal Otto cycle has the compression ratio 6. The initial pressure and temperature of air are 1 bar and 37 °C. The maximum pressure in the cycle is 30 bar. For unit mass flow, calculate (i) p, V and T at various salient points of the cycle and (ii) the ratio of heat supplied to the heat rejected. Assume $\gamma = 1.4$ and $R =$ 8.314 kJ/kmol K.

Solution

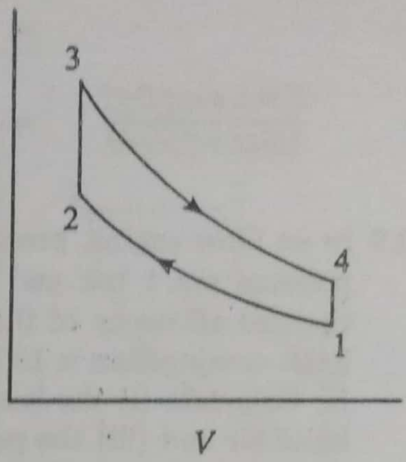
Consider point 1,

$$n = \frac{m}{M} = \frac{1}{29}$$

$$V_1 = \frac{nRT_1}{p_1} = \frac{1 \times 8314 \times 310}{29 \times 10^5} = \mathbf{0.889\ m^3} \quad \underleftarrow{\text{Ans}}$$

Consider point 2,

$$p_2 = p_1\ r^{\gamma} = 10^5 \times 6^{1.4} = 12.3 \times 10^5 \text{ N/m}^2 = \mathbf{12.3\ bar} \quad \underleftarrow{\text{Ans}}$$



$$V_2 = \frac{V_1}{6} = \frac{0.889}{6}$$

$$= 0.148 \text{ m}^3 \quad \stackrel{\text{Ans}}{\Longleftarrow}$$

$$T_2 = \frac{p_2 V_2}{p_1 V_1} T_1$$

$$= \frac{12.3 \times 10^5 \times 0.148 \times 310}{1 \times 10^5 \times 0.889}$$

$$= 634.8 \text{ K}$$

$$= 361.8^\circ \text{ C} \quad \stackrel{\text{Ans}}{\Longleftarrow}$$

Consider point 3,

$$V_3 = V_2 = 0.148 \text{ m}^3 \quad \stackrel{\text{Ans}}{\Longleftarrow}$$

$$p_3 = 30 \times 10^5 \text{ N/m}^2 = 30 \text{ bar} \quad \stackrel{\text{Ans}}{\Longleftarrow}$$

$$\frac{p_3}{T_3} = \frac{p_2}{T_2}$$

$$T_3 = \frac{30 \times 10^5}{12.3 \times 10^5} \times 634.8$$

$$= 1548 \text{ K} = 1275^\circ \text{ C} \quad \stackrel{\text{Ans}}{\Longleftarrow}$$

Consider point 4,

$$p_3 V_3^\gamma = p_4 V_4^\gamma$$

$$p_4 = p_3 \left(\frac{V_3}{V_4}\right)^\gamma = 30 \times 10^5 \left(\frac{1}{6}\right)^{1.4}$$

$$= 2.44 \times 10^5 \text{ N/m}^2 = 2.44 \text{ bar} \quad \stackrel{\text{Ans}}{\Longleftarrow}$$

$$V_4 = V_1 = 0.889 \text{ m}^3 \quad \stackrel{\text{Ans}}{\Longleftarrow}$$

$$T_4 = T_1 \frac{p_4}{p_1} = 310 \times \frac{2.44 \times 10^5}{1 \times 10^5}$$

$$= 756.4 \text{ K} = 483.4^\circ \text{ C} \quad \stackrel{\text{Ans}}{\Longleftarrow}$$

$$C_v = \frac{R}{M(\gamma - 1)} \frac{8.314}{29 \times 0.4} = 0.717 \text{ kJ/kg K}$$

For unit mass,

$$\textit{Heat supplied} = C_v(T_3 - T_2)$$

$$= 0.717 \times (1548 - 635.5) = 654.3 \text{ kJ}$$

$$\text{Heat rejected} = C_v(T_4 - T_1)$$

$$= 0.717 \times (756.4 - 310) = 320.1 \text{ kJ}$$

$$\frac{\text{Heat supplied}}{\text{Heat rejected}} = \frac{654.3}{320.1} = 2.04 \quad \underleftarrow{\text{Ans}}$$

2.9 In an Otto engine, pressure and temperature at the beginning of compression are 1 bar and 37 °C respectively. Calculate the theoretical thermal efficiency of this cycle if the pressure at the end of the adiabatic compression is 15 bar. Peak temperature during the cycle is 2000 K. Calculate (i) the heat supplied per kg of air (ii) the work done per kg of air and (iii) the pressure at the end of adiabatic expansion. Take $C_v = 0.717$ kJ/kg K and $\gamma = 1.4$.

Solution

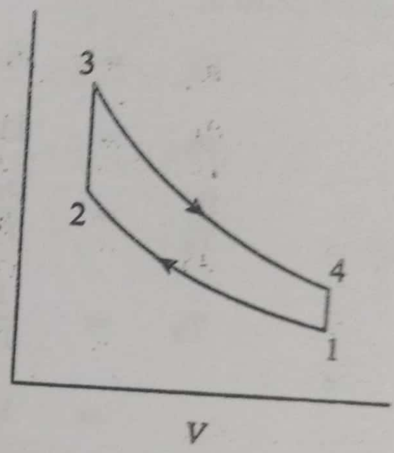
Consider the process 1-2

$$T_2 = T_1\left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$= 310 \times (15)^{(0.4/1.4)} = 672 \text{ K}$$

$$\eta = 1 - \frac{T_1}{T_2} = 1 - \frac{310}{672}$$

$$= 0.539 = 53.9\%$$



Consider the process 2 – 3

For unit mass flow,

$$q_s = C_v(T_3 - T_2)$$

$$= 0.717 \times (2000 - 672) = 952.2 \text{ kJ/kg} \quad \underleftarrow{\text{Ans}}$$

$$\eta = \frac{\text{Work done per kg of air}}{\text{Heat supplied per kg of air}} = \frac{w}{q_s}$$

$$w = \eta q_s = 0.539 \times 952.2 = 513.2 \text{ kJ/kg} \quad \underleftarrow{\text{Ans}}$$

$$p_3 = p_2\left(\frac{T_3}{T_2}\right) = 15 \times \frac{2000}{672} = 44.64 \text{ bar}$$

Consider the process 3 – 4

$$\frac{p_3}{p_4} = \left(\frac{V_4}{V_3}\right)^{\gamma} = \left(\frac{V_1}{V_2}\right)^{\gamma} = \frac{p_2}{p_1}$$

$$p_4 = p_3\left(\frac{p_1}{p_2}\right) = \frac{44.64 \times 1}{15} = 2.98 \text{ bar} \quad \underleftarrow{\text{Ans}}$$

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2.10 Compare the efficiencies of ideal Atkinson cycle and Otto cycle for a compression ratio is 5.5. The pressure and temperature of air at the beginning of compression stroke are 1 bar and 27 °C respectively. The peak pressure is 25 bar for both cycles. Assume $\gamma = 1.4$ for air.

Solution

$$\eta_{Otto} = 1 - \frac{1}{r^{\gamma-1}} = 1 - \frac{1}{5.5^{0.4}} = 49.43\%$$

Refer Fig.2.14

$$\eta_{Atkinson} = 1 - \frac{\gamma(e-r)}{e^{\gamma} - r^{\gamma}}$$

$$r = 5.5$$

$$e = \frac{V_{4'}}{V_3} = \left(\frac{p_3}{p_{4'}}\right)^{\frac{1}{\gamma}} = \left(\frac{25}{1}\right)^{\frac{1}{1.4}} = 9.966$$

$$e^{\gamma} = 9.966^{1.4} = 25$$

$$r^{\gamma} = 5.5^{1.4} = 10.88$$

$$\eta_{Atkinson} = 1 - \frac{1.4 \times (9.966 - 5.5)}{25 - 10.88} = 55.72\%$$

$$\frac{\eta_{Atkinson}}{\eta_{Otto}} = \frac{55.72}{49.43} = 1.127$$ $\Longleftarrow$ Ans

DIESEL CYCLE

2.11 A Diesel engine has a compression ratio of 20 and cut-off takes place at 5% of the stroke. Find the air-standard efficiency. Assume $\gamma = 1.4$.

Solution

$$r = \frac{V_1}{V_2} = 20$$

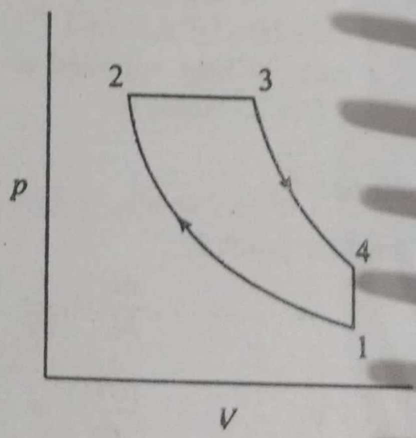
$$V_1 = 20V_2$$

$$V_s = 20V_2 - V_2 = 19V_2$$

$$V_3 = 0.05V_s + V_2$$

$$= 0.05 \times 19V_2 + V_2$$

$$= 1.95V_2$$



$$r_c = \frac{V_3}{V_2} = \frac{1.95V_2}{V_2} = 1.95$$

$$\eta = 1 - \frac{1}{r^{\gamma-1}}\ \frac{r_c^\gamma - 1}{\gamma(r_c - 1)}$$

$$= 1 - \frac{1}{20^{0.4}} \times \left[\frac{1.95^{1.4} - 1}{1.4 \times (1.95 - 1)}\right] = 0.649 = 64.9\% \quad \underleftarrow{\text{Ans}}$$

2.12 Determine the ideal efficiency of the diesel engine having a cylinder with bore 250 mm, stroke 375 mm and a clearance volume of 1500 cc, with fuel cut-off occurring at 5% of the stroke. Assume $\gamma = 1.4$ for air.

Solution

$$V_s = \frac{\pi}{4}\ d^2\ L = \frac{\pi}{4} \times 25^2 \times 37.5$$

$$= 18407.8 \text{ cc}$$

$$r = 1 + \frac{V_s}{V_c} = 1 + \frac{18407.8}{1500} = 13.27$$

$$\eta = 1 - \frac{1}{r^{\gamma-1}}\ \frac{r_c^\gamma - 1}{\gamma(r_c - 1)}$$

$$r_c = \frac{V_3}{V_2}$$

$$\text{Cut-off volume} = V_3 - V_2 = 0.05V_s = 0.05 \times 12.27V_c$$

$$V_2 = V_c$$

$$V_3 = 1.6135V_c$$

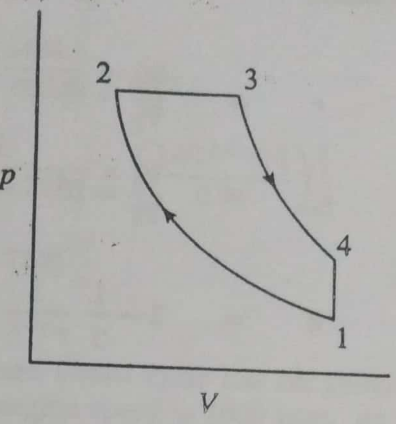
$$r_c = \frac{V_3}{V_2} = 1.6135$$

$$\eta = 1 - \frac{1}{13.27^{0.4}} \times \frac{1.6135^{1.4} - 1}{1.4 \times (1.6135 - 1)}$$

$$= 0.6052 = 60.52\% \quad \underleftarrow{\text{Ans}}$$

2.13 In an engine working on Diesel cycle inlet pressure and temperature are 1 bar and 17 °C respectively. Pressure at the end of adiabatic compression is 35 bar. The ratio of expansion i.e. after constant pressure heat addition is 5. Calculate the heat addition, heat rejection and the efficiency of the cycle. Assume $\gamma = 1.4$, $C_p = 1.004$ kJ/kg K and $C_v = 0.717$ kJ/kg K.

Solution



Consider the process 1 – 2

$$\frac{V_1}{V_2} = r$$

$$= \left(\frac{p_2}{p_1}\right)^{\frac{1}{\gamma}}$$

$$= \left(\frac{35}{1}\right)^{\frac{1}{1.4}} = 12.674$$

$$r_c = \frac{V_3}{V_2} = \frac{V_3}{V_1} \times \frac{V_1}{V_2}$$

$$= \frac{\text{Compression ratio}}{\text{Expansion ratio}} = \frac{12.674}{5} = 2.535$$

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{35}{1}\right)^{0.286} = 2.76$$

$$T_2 = 2.76 \times 290 = 801.7 \text{ K}$$

Consider the process 2 – 3

$$T_3 = T_2\frac{V_3}{V_2} = 801.7 \times \frac{V_3}{V_2} = 801.7 \times 2.535 = 2032.3 \text{ K}$$

Consider the process 3 – 4

$$T_4 = T_3\left(\frac{V_3}{V_4}\right)^{\gamma-1} = 2032.3 \times \left(\frac{1}{5}\right)^{0.4} = 1067.6 \text{ K}$$

$$\text{Heat added} = C_p(T_3 - T_2) = 1.004 \times (2032.3 - 801.7)$$

$$= \mathbf{1235.5\ kJ/kg}$$ Ans

$$\text{Heat rejected} = C_v(T_4 - T_1) = 0.717 \times (1067.6 - 290)$$

$$= \mathbf{557.5\ kJ/kg}$$ Ans

$$\text{Efficiency} = \frac{\text{Heat supplied - Heat rejected}}{\text{Heat supplied}}$$

$$= \frac{1235.5 - 557.5}{1235.5} = 0.549 = 54.9\%$$ Ans